

engineer, Hugo Leuteritz, developed a long-range radio direction finder to supplement traditional navigation methods. The aircraft transmitted a signal that allowed a ground station on one of the islands to establish its position, which was transmitted back to the aircraft's navigator.

Pacific crossings

Equipped with the new radio-navigation equipment, the S-42 with Pan American's star pilot Ed Musick at the controls set off from San Francisco in April 1935 on the first experimental flight to Hawaii. To maximize its range, the S-42 had been stripped of all superfluous weight and packed with extra fuel tanks. Even so, disaster was only narrowly averted. The outward flight went smoothly, reaching Honolulu in 18 hours 37 minutes. But on the return journey Musick had to battle headwinds, which added five hours to the flight time – putting it far beyond the S-42's theoretical endurance limit of 21 hours. By some miracle, there was still a little fuel in the tanks when the airplane reached California. Despite such teething troubles, the S-42 survey flights were

successfully completed, and on November 22, 1935, the Martin 130 *China Clipper* set out from San Francisco on the first scheduled Pacific airmail flight to Manila in the Philippines. It was given a tumultuous sendoff and an equally tumultuous reception at its destination, where more than 300,000 people gathered.

Passenger flights had to wait until crews had built up more experience of flying the route, and until hotels had been built at the island stopovers. But finally, on October 21, 1936, the Martin 130 *Philippine Clipper* set off from California with 15 passengers on board, arriving in Hong Kong three days later. The same journey by boat usually took three weeks.

Of course, the Americans had no monopoly over flying-boat operations. In the 1930s, France's Latécoère flying boats

operated in the Mediterranean and across the South Atlantic, while from 1934 Lufthansa flew a scheduled service from Berlin to Rio de Janeiro via Spain and West Africa. Britain used flying boats on its imperial routes to South Africa and Australia. At first these journeys required a mix



FINE DINING

The Sikorsky S-42 Clipper was equipped with a galley, allowing Pan American to serve high-quality meals to its passengers. In addition to serving refreshments, it was also the steward's job to point out scenic attractions through the windows.



VARIETY OF DESTINATIONS

The cover of a Pan American timetable advertises a wide range of destinations. Regular passenger services between the US and the Philippines started in October 1936, charging passengers \$799 each for the 8,200-mile (13,200-km) flight.

